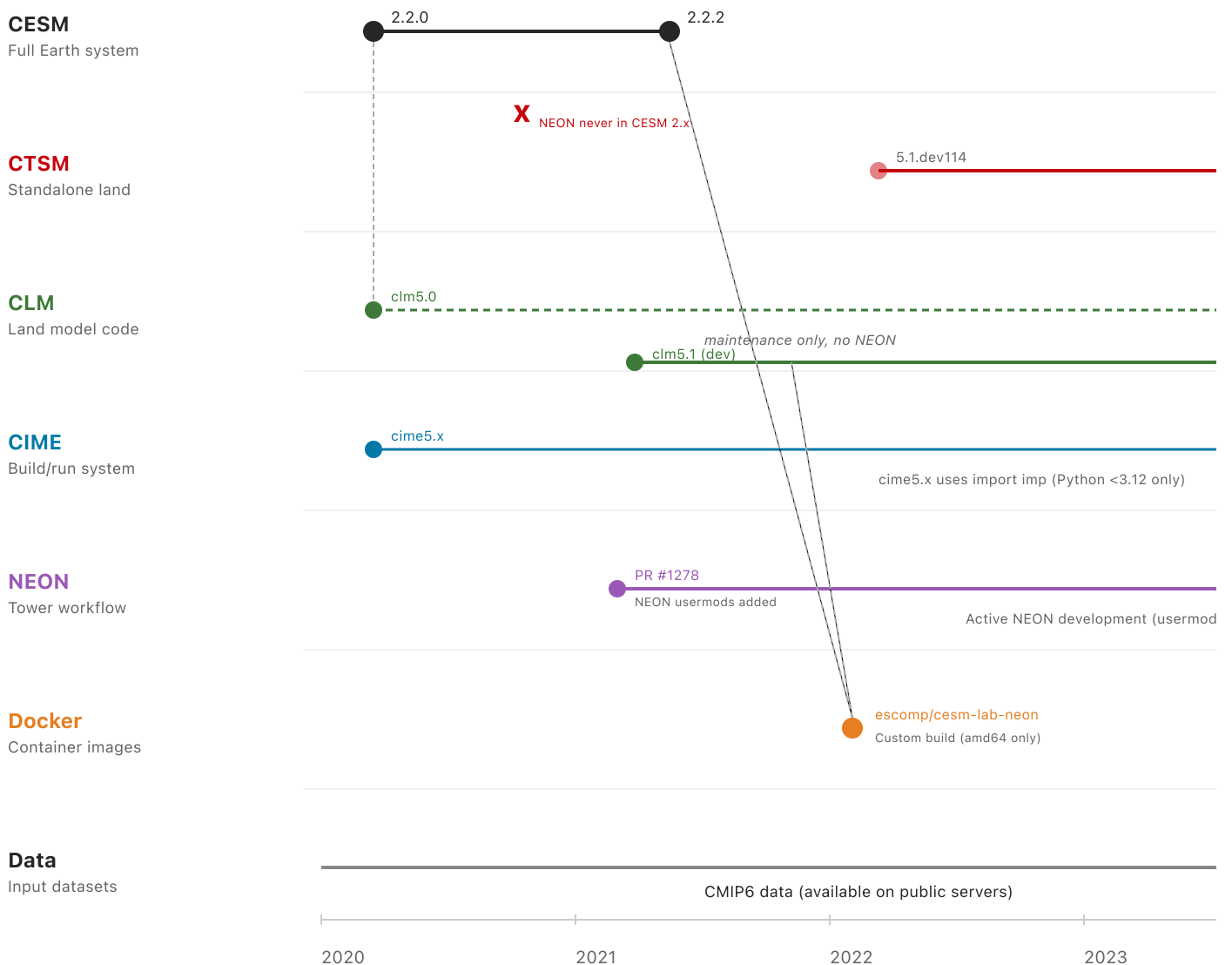


CESM / CTSM Component Version Lineage

How the components, versions, and releases connect over time



● CESM (full coupled Earth system) ● CTSM (standalone land model) ● CLM (land model code) ● CIME (build/run infrastructure) ● NEON (tower site workflow) ● Docker (container images)

Solid lines = released. Dashed lines = maintenance/beta/partial. Large filled circles = current/key releases. Lines connecting rows show composition (what contains what).

Key takeaways from this diagram:

- NEON support (purple) was developed on the CTSM development branch starting Feb 2021, never on the CESM 2.x line.
- The old escomp/cesm-lab-neon Docker image (2022) was a custom build mixing CESM 2.2 with a dev CTSM branch (dashed lines).
- CLM has two branches: clm5.0 (maintenance, used by CESM 2.2, no NEON) and clm5.1/6.0 (active development, used by CTSM).

- CIME upgraded from 5.x (Python <3.12 only) to 6.x (Python 3.12+) between CTSM 5.2 and 5.4.
- CMIP7 input data (red dashed line at bottom) is only partially published, blocking live simulations with CTSM 5.4.
- The ExSOIL container uses CTSM 5.4, which includes CLM 6.0, CIME 6.1, and 48 NEON sites natively.